

MJOLFJELL, Norway

WMO: 013440

Lat: 60.702N

Lon: 6.937E

Elev: 695

StdP: 93.25

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.9	-15.2	-20.7	0.6	-14.2	-19.2	0.7	-13.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	22.7	14.0	20.4	13.0	18.2	11.9	14.8	20.8	13.8	19.0	12.8	17.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	9.7	16.2	11.4	9.1	15.4	10.5	8.6	14.6	43.6	21.1	40.9	19.3	38.2	17.4	18.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-18.5	25.0	1.5	2.2	-19.6	26.6	-20.5	27.9	-21.3	29.1	-22.4	30.7
			-19.0	15.9	1.5	1.6	-20.1	17.1	-20.9	18.0	-21.8	19.0	-22.8	20.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-3.9	-3.9	-1.2	2.2	6.1	9.5	12.7	11.7	8.4	3.9	0.2	-2.2
	DBStd	6.87	4.57	5.08	3.92	3.62	3.49	2.71	3.38	2.65	3.06	3.43	3.98	4.46
	HDD10.0	2523	432	390	346	235	129	41	11	12	66	189	293	379
	HDD18.3	5352	690	623	604	484	378	264	177	207	297	446	543	638
	CDD10.0	214	0	0	0	0	9	27	95	64	19	1	0	0
	CDD18.3	3	0	0	0	0	0	0	3	0	0	0	0	0
	CDH23.3	43	0	0	0	0	1	0	41	2	0	0	0	0
	CDH26.7	3	0	0	0	0	0	0	3	0	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	1714	208	142	153	82	75	87	101	122	152	150	165	208
	PrecMax	2400	409	370	540	173	151	174	156	189	263	314	328	428
	PrecMin	1179	21	0	28	33	17	25	44	54	51	33	18	12
	PrecStd	314	109	98	113	36	39	40	30	35	63	79	88	117
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	6.2	9.0	13.7	21.4	21.6	26.9	23.0	19.1	14.0	10.6	6.5
		MCWB	2.5	3.3	3.7	6.4	11.8	12.4	15.6	14.4	11.8	9.2	6.5	4.4
	2%	DB	3.6	4.3	6.2	11.1	17.4	19.1	24.3	20.4	16.9	11.6	8.5	5.1
		MCWB	1.8	2.3	2.6	5.6	10.2	10.9	14.7	13.5	11.4	8.4	5.5	3.4
	5%	DB	2.7	3.0	4.5	9.8	15.0	17.2	22.0	18.3	14.8	10.1	6.6	3.9
		MCWB	1.1	1.3	1.9	5.1	8.9	10.2	14.0	12.4	10.5	7.1	4.4	2.3
	10%	DB	1.6	2.0	3.4	8.3	12.8	15.6	19.3	16.7	13.4	8.7	5.3	2.9
		MCWB	0.3	0.4	1.2	4.2	7.7	9.4	13.0	11.6	9.7	6.0	3.3	1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.9	4.4	7.9	12.3	13.4	16.7	15.2	13.6	10.9	7.2	4.8
		MCDB	4.0	5.4	7.1	11.8	19.5	19.2	24.0	20.8	17.2	12.7	9.6	5.8
	2%	WB	2.2	2.6	3.3	6.4	10.6	12.2	15.4	14.2	12.1	8.7	6.0	3.7
		MCDB	3.1	3.9	5.2	10.3	16.4	17.7	22.3	19.2	15.3	10.9	7.8	4.9
	5%	WB	1.3	1.7	2.5	5.2	9.3	11.2	14.5	13.2	11.0	7.7	4.8	2.6
		MCDB	2.4	2.9	4.1	9.3	14.5	15.8	20.5	17.3	14.3	9.9	6.4	3.6
	10%	WB	0.5	0.7	1.5	4.2	8.0	10.1	13.5	12.3	10.0	6.4	3.4	1.6
		MCDB	1.5	1.7	3.1	7.9	12.3	14.1	18.7	15.6	12.9	8.4	5.0	2.6
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.8	6.9	8.0	8.5	8.9	8.5	7.5	6.5	5.6	4.7	5.0
		MCDBR	5.1	4.7	7.4	10.0	12.4	12.8	13.3	10.5	9.6	7.3	5.8	5.0
		MCWBR	4.2	3.7	4.7	5.6	6.4	6.2	5.2	4.8	5.1	4.7	4.3	4.2
	5% WB	MCDBR	4.9	4.8	6.7	9.1	11.6	10.8	11.6	9.5	8.4	6.1	5.6	5.1
		MCWBR	4.2	4.0	4.7	5.2	6.3	5.8	4.9	4.9	4.7	4.6	4.6	4.5
Clear-Sky Solar Irradiance	taub	0.222	0.264	0.296	0.329	0.334	0.331	0.344	0.336	0.309	0.290	0.234	0.194	
	taud	2.106	2.231	2.202	2.180	2.327	2.443	2.435	2.450	2.516	2.442	2.166	1.989	
	Ebn at Noon	543	737	825	862	889	899	876	855	823	711	518	401	
	Edh at Noon	39	66	95	118	111	101	100	91	71	54	36	27	
All-Sky Solar Radiation	RadAvg	0.24	0.76	2.02	3.84	4.79	4.79	4.59	3.56	2.25	1.04	0.34	0.16	
	RadStd	0.06	0.22	0.40	0.65	0.80	0.56	0.56	0.33	0.27	0.15	0.05	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page